

Yaowen Zhang

yaowenzhang196@outlook.com | [GitHub](#) | [Personal Website](#) | [Google Scholar](#)

Research Interests: Multimodal Learning; Vision-Language Models; Foundation Models; Computer Vision

EDUCATION

Nanyang Technological University (NTU), Singapore 2024–2026
M.Sc. in Artificial Intelligence, College of Computing and Data Science
GPA: 4.20/5.00

University of Electronic Science and Technology of China (UESTC), Chengdu, China 2020–2024
B.Eng. in Computer Science and Technology
GPA: 87.84/100 (3.88/4.00)
Undergraduate Thesis: *Pedestrian Detection via Transfer Learning*

PUBLICATIONS

Yaowen Zhang, Xiangyu Qiu, Junyi Hu, Zhi Lu, Wenwen Tian, Aoqin Wang, Junhai Luo, Zhenming Peng.
ReSight-SMC: Two-Stage Power Sampling via Island SMC with Visual Scouts.
Preprint, arXiv:2609.34905, 2026. Under review at ICLR 2027. [\[Paper\]](#) [\[Code\]](#)

Junyi Hu, Fengyi Wu, Tian Bai, Yian Huang, Xiangyu Qiu, **Yaowen Zhang**, Junli Lin, Zhenming Peng, Yi Zhang.
Language-Conditioned Mean-Field Unfolding: A Variational Inference Paradigm for Referring Expression Segmentation.
Under review at ICLR 2027.

RESEARCH PROJECTS

ReSight-SMC — First author 2026.6–2026.9

- Developed **ReSight-SMC**, a verifier-free two-stage test-time scaling framework for LVLMs that combines ancestry-isolated island SMC, prefix-conditioned visual scouts, and answer-level power readout to improve finite-particle exploration and decision aggregation.
- Designed prefix-conditioned visual routing that sends a bounded subset of particles to complementary image regions, reactivates visual attention during decoding, and uses exact importance correction to preserve the base LVLM sequence-power target; provided theoretical analysis for the corrected adaptive proposals.
- Led problem formulation, method design, implementation, experiments, and manuscript preparation; evaluated four LVLM backbones across five benchmarks, achieving stronger aggregate performance than Power-SMC on both reasoning and perception benchmark groups while remaining competitive with backbone-matched RL models without post-training.

Memory-Efficient SpatialRGPT — Individual Course Project (3D Deep Learning) 2025.4–2025.5

- Replaced SpatialRGPT's depth estimator with Depth Anything V2; used xFormers and capped depth-input width to reduce inference memory without retraining.
- Reduced GPU memory use from over 48 GB to 22.7 GB, enabling inference on a 24 GB GPU; raised SpatialRGPT-Bench qualitative accuracy from 92.69% to 93.61%.

Reducing Object Hallucination in ViP-LLaVA — Team Lead, Deep Learning Course Project 2024.10–2024.11

- Led a five-person team; implemented the ViP-LLaVA-7B adaptation of visual contrastive decoding and answer-format prompts, and ran POPE and ViP-Bench evaluations.
- Improved POPE accuracy in all nine settings (A-OKVQA adversarial: 69.4% to 78.0%) and ViP-Bench bounding-box score from 41.8 to 45.4. [\[Code\]](#)

PROFESSIONAL EXPERIENCE

Independent Researcher 2026–Present

- Conduct self-directed research on vision-language model reasoning and test-time scaling; lead research questions, method development, implementation, evaluation, and manuscript writing.

JD.com (Joybuy) — Software Development Engineer 2026–Present

- Developed front-end features and backend-for-frontend (BFF) services for Joybuy's OPC system, supporting JD.com's European online retail business.
- Developed business-facing AI skills for OPC workflows.

HONORS AND AWARDS

Undergraduate Scholarship (2024)

TECHNICAL SKILLS

Programming: Python, TypeScript, Java
ML / Systems: PyTorch, Hugging Face, xFormers, LangChain, CUDA, Slurm, Docker, Linux, Git